PORTFOLIO CASE STUDY: Enterprise Intelligent Automation & SAP Workflow Orchestration

- **Platform:** Tungsten TotalAgility (TA), Process Director Accounts Payable, SAP ERP Integration
- **Role:** Senior Business Analyst
- **Domain:** Fintech / Intelligent Document Processing (IDP)

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1. Context & Business Challenge

A client (ABC) handling 200,000+ unstructured multi-format documents quarterly across international logistics faced numerous operational risks due to manual invoice validation and paper-heavy workflows.

- **Bottlenecks:** Accounts Payable (AP) teams were spending excessive manual effort reworking header data, tax details, and line items, causing an average processing cycle time of 12 days.
- **Compliance & Visibility Risks:** An inability to maintain systematic data led to loss in early-payment incentives and supplier relationship strain.

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2. BA Elicitation & Solution Strategy

As the Lead Business Analyst, I served as the core functional bridge between Finance stakeholders, Compliance units, and the Technical Delivery team to replace the legacy procedure with an integrated deployment of **Tungsten TotalAgility (IDP Engine)** and **Process Director (SAP Workflow Layer)**.

- **Elicitation:** Communicated with the clint in ER that was raised, executed process-shadowing sessions on calls and stakeholder workshops to isolate validation gaps.
- **Process Transformation:** Defined the transition from template-dependent extraction models to an **AI-driven touchless workflow** leveraging TotalAgility's pre-trained cognitive asset libraries.

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3. Core Portfolio Artifacts Delivered

Artifact A: Requirements Documentation Framework (BRD vs. FRD)
To ensure clear alignment across both business stakeholders and technical developers, I authored and managed two primary specification blueprints for this implementation:

BRD
This document answered the question: "What does the business need to achieve?" It was written in clear business language for senior leadership and finance executives.
Key Sections:

• Current-State (As-Is): Documented the manual entry workflows causing the processing bottleneck.

• Future-State (To-Be) Objectives: Defined the target goal of achieving a 48-hour processing window and cutting processing costs.

• High-Level Scope Boundary: Established what types of documents were included (e.g., standard PDF vendor invoices) and what was explicitly out of scope (e.g., manual paper receipts).

• Success Metrics (KPIs): Defined the target of achieving a 65% Straight-Through Processing (STP) rate post-launch.

FRD
This document answered the question: "How must the system behave to meet those business needs?" It translated the business goals into specific technical rules for the Tungsten and SAP development teams.
Key Sections:

- System Ingestion Rules: Specified exactly how the system must pull files from automated email inboxes and dedicated SFTP folders.
- Data Validation Logic: Outlined the strict 3-way matching rules required to cross-verify Invoice totals against active SAP Purchase Orders and Goods Receipts.
- Confidence Threshold Parameters: Defined the specific rule where the product must achieve a ≥ 95% extraction confidence score before sending data to SAP without human review.
- Exception Error-Handling Workflows: Detailed the exact layout interface and routing paths required when a document failed validation or fell below accuracy thresholds.

Artifact B Target Workflows & System Mapping

- Ingestion Layer: Inbound PDF/Image assets mapped via multi-channel entry points into the TotalAgility Cognitive Capture Engine.
- Reconciliation Flow: Data streams are pushed systematically into Process Director to perform automated 3-Way Matching Rules (Comparing Vendor Invoice vs. SAP Purchase Order vs. Goods Receipt Records).

Artifact B: Agile Epics & User Stories

- **Epic: Data Extraction**
    - **User Story 1.1: Automated Field Extraction**
        - As a Lead Accounts Payable Processor, I want TotalAgility to parse incoming unstructured documents and isolate *Vendor Tax IDs, Line Item Totals, PO References,* and *Currency Identifiers* without static system templates, so that manual data entry tasks are eliminated.
        - **Acceptance Criteria:**
            - The product must match invoice data with at least 95% accuracy to pass it straight to SAP without a human check.
    - **User Story 1.2: Dynamic Exception Routing Blueprint**
        - As an Audit & Compliance Supervisor, I want extraction instances falling below the 95% threshold to automatically trigger an exception state and route to a designated human verification queue, so that accurate data-integrity checks are maintained prior to database commits.
        - **Acceptance Criteria:**
            - Given an extracted document metadata batch flags a variance check error or low extraction confidence,
            - When the workflow orchestration phase executes,
            - Then route the complete functional envelope to the validation layout space with high-priority status and clear mismatch field indicators.

Artifact C: Data Exchange Logic Mapping

What TotalAgility reads	SAP	Validation Check
Vendor Tax ID	Vendor Profile	Clean up the text spacing and make sure the tax ID format matches the vendor's country rules.
Purchase Order (PO) Number	Purchase Order Record	Check the database to ensure the PO is active and still has enough money left on it to pay the invoice.
Line-item Net Price	Accounting Ledger	Add up all the individual line items to make sure they equal the total amount at the top of the invoice (allowing a tiny variance of ±$0.02 for rounding errors).

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4. Measurable Project Business Impact

The implementation successfully optimized the core enterprise document pipeline within 6 months of release deployment:

- Reduced end-to-end processing cycles by 60%, dropping historical wait timelines down to a 48-hour completion window.

- Attained a 65% Touchless Straight-Through Processing (STP) rate on structured formats, reducing processing costs significantly.
- Established fully automated digital audit trails, completely removing manual documentation reconciliation tracking efforts and validation data gaps.